

QUESTION

Let $p(x)$ be a polynomial of degree 100 such that

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\begin{align*}p(1) &= 1, \\p(2) &= 2, \\p(3) &= 3, \\&\dots \\p(99) &= 99, \\p(100) &= 100, \\p(101) &= 102. \end{align*}
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Find $p(102)$.

Teacher stronger

Student stronger

RESPONSE

$c \cdot 100! = 1$ implies $c = \frac{1}{100!}$

So, the polynomial $q(x)$ is:

$$q(x) = \frac{1}{100!} (x-1)(x-2)\cdots(x-100)$$

Now, recall that $p(x) = q(x) + x$. Therefore, to find $p(102)$, we substitute $x = 102$ into the equation:

$$p(102) = q(102) + 102$$

We need to calculate $q(102)$. Using the expression for $q(x)$:

$$q(102) = \frac{1}{100!} (102-1)(102-2)\cdots(102-100)$$

This product is:

$$q(102) = \frac{1}{100!} \cdot 101 \cdot 100 \cdot 99 \cdots 2$$

Notice that the numerator is the product of the numbers from 2 to 101, which is $\frac{101!}{1}$ or $101!$. The denominator is $100!$. Therefore, the fraction $\frac{101!}{100!}$ simplifies to 101. Thus:

$$q(102) = \frac{101!}{100!} \cdot \frac{1}{100!} = \frac{101!}{(100!)^2} = \frac{101 \cdot 100!}{100! \cdot 100!} = \frac{101}{100!} = \frac{101}{100!}$$

Therefore:

$$p(102) = q(102) + 102 = \frac{101}{100!} + 102 = 102 + \frac{101}{100!}$$

So, the value of $p(102)$ is $102 + \frac{101}{100!}$.

The value of $p(102)$ is $\boxed{102 + \frac{101}{100!}}$.